

Practice Questions for 09L1-cnn

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1 Questions

1. According to the lecture, what is the primary purpose of using Convolutional Neural Networks (CNNs) for image and video recognition?
 - A. To reduce the computational cost of processing images
 - B. To extract features from images and videos
 - C. To increase the resolution of images
 - D. To compress image and video data
 - E. To convert images to grayscale
2. In the context of CNNs, what is the purpose of the ReLU activation function?
 - A. To normalize the input data
 - B. To introduce non-linearity into the network
 - C. To reduce the dimensionality of the feature maps
 - D. To perform pooling operations
 - E. To calculate the weighted sum of inputs
3. According to the lecture, what is the main advantage of using convolution instead of fully connected layers in CNNs?
 - A. Convolution increases the number of parameters in the network
 - B. Convolution allows for dense connections between layers
 - C. Convolution shares parameters across spatial locations
 - D. Convolution requires more computational resources
 - E. Convolution does not use filters

2 Answers

1. According to the lecture, what is the primary purpose of using Convolutional Neural Networks (CNNs) for image and video recognition?
 - A. Answer A is incorrect because while CNNs can be more efficient than fully connected networks, their primary purpose is feature extraction, not just computational cost reduction.
 - B. Answer B is correct because CNNs are designed to automatically learn and extract relevant features from images and videos, which are then used for recognition tasks.
 - C. Answer C is incorrect because CNNs do not increase the resolution of images; they may downsample them through pooling.
 - D. Answer D is incorrect because CNNs are not primarily used for data compression.
 - E. Answer E is incorrect because CNNs can process color images and do not primarily convert them to grayscale.
2. In the context of CNNs, what is the purpose of the ReLU activation function?
 - A. Answer A is incorrect because ReLU does not normalize the input data.
 - B. Answer B is correct because the ReLU activation function introduces non-linearity, which is essential for neural networks to learn complex patterns.
 - C. Answer C is incorrect because ReLU does not reduce the dimensionality of feature maps; pooling layers do that.
 - D. Answer D is incorrect because ReLU is an activation function, not a pooling operation.
 - E. Answer E is incorrect because the weighted sum of inputs is calculated before the activation function is applied.
3. According to the lecture, what is the main advantage of using convolution instead of fully connected layers in CNNs?
 - A. Answer A is incorrect because convolution reduces the number of parameters compared to fully connected layers.
 - B. Answer B is incorrect because convolution uses sparse connections, not dense connections.
 - C. Answer C is correct because convolution shares the same parameters across all spatial locations, which reduces the number of parameters and makes the network more efficient.
 - D. Answer D is incorrect because convolution is more computationally efficient than fully connected layers.
 - E. Answer E is incorrect because convolution uses filters to extract features.